meta-analysis for ammonia emission after deep-placement

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Table of contents

1 package import

```
library(metafor)
library(dplyr)
library(ggplot2)
library(orchaRd)
library(patchwork)
library(gt)
```

2 data import

```r
# read data ------------------------------------------------------------------
dat <- readxl::read_excel("sub-placement-2.xlsx", 1)

# prepare ---
dat2 <- dat |>
  filter(mea_env == "field") |>
  # filter(crop != "sugarcane") |>
  mutate(deep_loss = ifelse(deep_loss == 0,
                            deep_loss + 0.001,
                            deep_loss)) |>
  mutate(sub_depth2 = cut(sub_depth,
                          breaks = c(0, 5, 10, 50),
                          labels=c("<5", "5~10", ">10"))) |>
  mutate(soil_clay2 = cut(soil_clay,
                          breaks = c(0, 18, 35, 100),
                          labels=c("<18%", "18%~35%", ">35%"))) |>
  mutate(SOC2 = cut(SOC,
                    breaks = c(0, 1, 2, 10),
                    labels = c("<1%", "1%~2%", ">2%"))) |>
  mutate(soil_pH2 = cut(soil_pH,
                        breaks = c(0, 6, 7, 8, 10),
                        labels = c("<6", "6~7","7~8",">8"))) |>
  mutate(
    n1i = rep_N, n2i = rep_N,
    sd_deep = deep_se * sqrt(n2i), # from se to sd
    sd_surface = surface_se * sqrt(n1i) # from se to sd
  )
```

3 calculate effect size

```r
es <- escalc(
  measure = "ROM",
  m1i = deep_loss, sd1i = sd_deep, n1i = n1i,
  m2i = surface_loss, sd2i = sd_surface, n2i = n2i,
  data = dat2,
  append = TRUE
)
```

4 fit a overall model

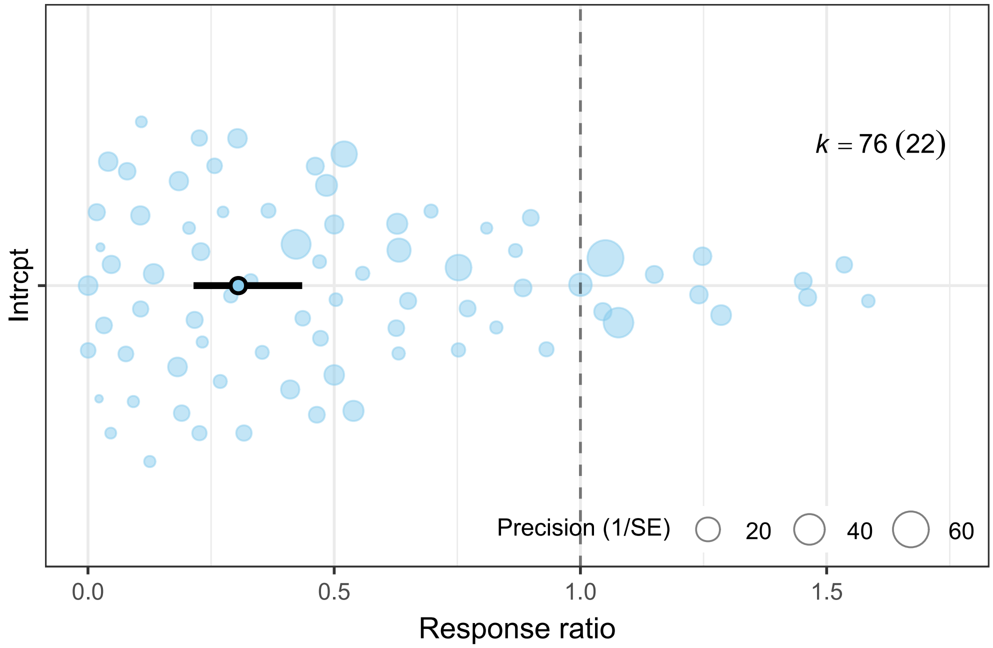
```r
res0 <- rma(yi, vi, data = es)

res0
```

```
Random-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 2.3147 (SE = 0.3996)
tau (square root of estimated tau^2 value):      1.5214
I^2 (total heterogeneity / total variability):   99.65%
H^2 (total variability / sampling variability):  285.89

Test for Heterogeneity:
Q(df = 75) = 11788.8270, p-val < .0001

Model Results:

estimate      se     zval    pval    ci.lb    ci.ub
 -1.1864  0.1806  -6.5701  <.0001  -1.5403  -0.8324  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_res0 <- res0 |>
  mod_results(group="ref") |>
  orchaRd:::transform_mod_results(transfm="exp")

mod_res0$mod_table |>
  mutate(across(where(is.numeric),\(x) round(x , digit=3)))
```

```
     name estimate lowerCL upperCL lowerPR upperPR
1 Intrcpt    0.305   0.214   0.435   0.015    6.15
```

```
mod_res0 |>
  orchard_plot(xlab="Response ratio",
               twig.size=0, refline.pos=1)
```



5 build a function for sub-group analysis

```
sub_meta <- function(dat, mod = "1", group = "ref", method = "rma",
                     transfm = "none", refline.pos = 1,
                   es.lab = "Response ratio", angle = 0, point.col = "#C93756")
{

  mods <- paste("~", mod, "-1") |>
    as.formula()

  random <- paste("~", "1", "|", group) |>
    as.formula()

  if (method == "mv") {
    res_sub <- rma.mv(yi, vi,
      mods = mods,
      random = random,
      data = dat
    )
  } else {
    res_sub <- rma(yi, vi,
      mods = mods,
      data = dat
    )
  }

  mod_sub <- res_sub |>
```

```r
    mod_results(group = group, mod = mod) |>
    orchaRd:::transform_mod_results(transfm = transfm)

  mod_table <- mod_sub$mod_table |>
    mutate(across(where(is.numeric), \(x) round(x, digit=3)))

  if (is.numeric(dat[[mod]])) {
    fig_sub <- mod_sub |>
      bubble_plot(
        group = group, g = T, pi.lwd = 0, pi.col = NA,
        xlab = mod, ylab = es.lab, est.col = point.col
      )  +
      labs(title = paste("Meta regression of", es.lab, "on", mod))

    out <- res_sub$data |>
      correlation::cor_test(x=mod, y="yi")
  } else {
    fig_sub <- mod_sub |>
      orchard_plot(xlab = es.lab, twig.size = 0,
                   refline.pos = refline.pos,
        angle = angle
      ) +
      labs(title = paste("Subgroup analysis of", mod))

    coefs  <- coef(res_sub)
    VC     <- vcov(res_sub)
    levels <- names(coefs)

  # build all pairwise combinations
  pairs <- combn(levels, 2, simplify = FALSE)

  results <- lapply(pairs, function(pr) {
    i <- pr[1]
    j <- pr[2]
    diff <- coefs[i] - coefs[j]
    se   <- sqrt(VC[i,i] + VC[j,j] - 2*VC[i,j])
    z    <- diff / se
    p    <- 2 * (1 - pnorm(abs(z)))
    tibble::tibble(level1 = stringr::str_remove(i, mod),
                   level2 = stringr::str_remove(j, mod),
               diff = diff, se = se, z = z, p = p)
  })

  out <- do.call(rbind, results)

  }

   res <- list(mod=mod, model = res_sub, comp=out, mod_res = mod_table, fig =
```

```
fig_sub)

  return(res)
}
```

6 build sub group meta-analysis for each moderator

6.1 factor list

- crop, season, country
- N_type, rate
- sub_tech, sub_type, sub_depth, sub_depth2
- soil_texture, soil_texture2
- soil_clay, soil_clay2
- soil_pH, soil_pH2
- SOC, SOC2
- SWC_w, SW_condition
- air_temp, rainfall
- mea_method

7 Moderator analysis

For each element in the variable vector, the meta-analysis will be done separately. The result includes, you can use "$element_name" to check:

- moderator name "mod"
- model "model"
- confidence interval data frame "mod_res"
- orchard or bubble plot "fig"

7.1 crop type

```
mod_crop <- sub_meta(dat = es, mod = "crop", transfm = "exp",
  es.lab="Response ration (RR)", refline.pos = 1)

mod_crop$model
```

```


Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2135 (SE = 0.3972)
tau (square root of estimated tau^2 value):             1.4878
I^2 (residual heterogeneity / unaccounted variability): 99.52%
H^2 (unaccounted variability / sampling variability):   207.85

Test for Residual Heterogeneity:
```

```
QE(df = 70) = 6818.2659, p-val < .0001

Test of Moderators (coefficients 1:6):
QM(df = 6) = 53.7949, p-val < .0001

Model Results:

               estimate      se     zval    pval    ci.lb    ci.ub
cropbamboo      -0.5998  0.8592  -0.6981  0.4851  -2.2839   1.0843
cropbare        -0.9507  0.3162  -3.0065  0.0026  -1.5705  -0.3309   **
cropcotton       0.1882  0.7451   0.2526  0.8006  -1.2721   1.6485
cropmaize       -1.2529  0.2884  -4.3447  <.0001  -1.8182  -0.6877  ***
cropsugarcane   -0.9425  1.0543  -0.8939  0.3714  -3.0088   1.1239
cropwheat       -2.0335  0.4106  -4.9526  <.0001  -2.8382  -1.2288  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_crop$comp
```

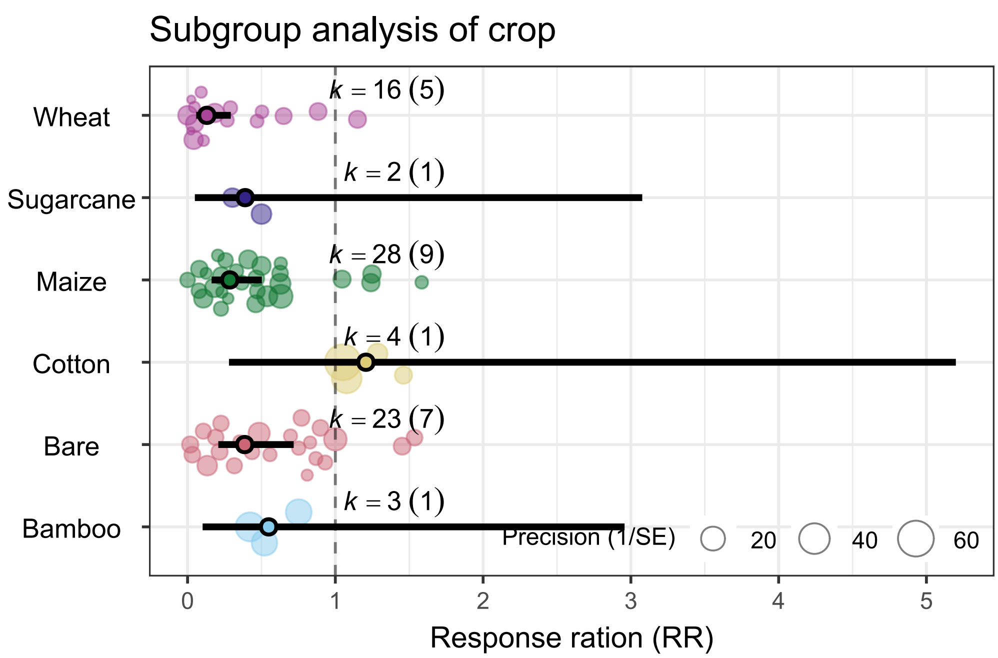
```
# A tibble: 15 × 6
   level1    level2        diff    se        z       p
   <chr>     <chr>        <dbl> <dbl>    <dbl>   <dbl>
 1 bamboo    bare       0.351   0.916  0.383   0.702
 2 bamboo    cotton    -0.788   1.14  -0.693   0.488
 3 bamboo    maize      0.653   0.906  0.721   0.471
 4 bamboo    sugarcane  0.343   1.36   0.252   0.801
 5 bamboo    wheat      1.43    0.952  1.51    0.132
 6 bare      cotton    -1.14    0.809 -1.41    0.159
 7 bare      maize      0.302   0.428  0.706   0.480
 8 bare      sugarcane -0.00826 1.10  -0.00750 0.994
 9 bare      wheat      1.08    0.518  2.09    0.0367
10 cotton    maize      1.44    0.799  1.80    0.0712
11 cotton    sugarcane  1.13    1.29   0.876   0.381
12 cotton    wheat      2.22    0.851  2.61    0.00901
13 maize     sugarcane -0.310   1.09  -0.284   0.776
14 maize     wheat      0.781   0.502  1.56    0.120
15 sugarcane wheat      1.09    1.13   0.964   0.335
```

```
mod_crop$mod_res
```

```
     name estimate lowerCL upperCL lowerPR upperPR
1    Bamboo    0.549   0.102   2.957   0.019  15.919
2      Bare    0.386   0.208   0.718   0.020   7.617
```

```
3     Cotton     1.207    0.280    5.199    0.046   31.482
4      Maize     0.286    0.162    0.503    0.015    5.570
5  Sugarcane     0.390    0.049    3.077    0.011   13.894
6      Wheat     0.131    0.059    0.293    0.006    2.695
```

```
mod_crop$fig
```



7.2 country

```
mod_country <- sub_meta(dat = es, mod = "country", transfm = "exp",
  es.lab="Response ration (RR)", refline.pos = 1)

mod_country$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     1.9795 (SE = 0.3653)
tau (square root of estimated tau^2 value):             1.4069
I^2 (residual heterogeneity / unaccounted variability): 99.49%
H^2 (unaccounted variability / sampling variability):   194.23

Test for Residual Heterogeneity:
QE(df = 67) = 5822.3667, p-val < .0001
```

```
Test of Moderators (coefficients 1:9):
QM(df = 9) = 68.9623, p-val < .0001

Model Results:

                    estimate      se     zval    pval    ci.lb    ci.ub
countryArgentina     -0.5416  0.9965  -0.5435  0.5868  -2.4948   1.4116
countryAustralia     -2.9172  0.7198  -4.0527  <.0001  -4.3280  -1.5064  ***
countryBrazil        -2.5803  0.5352  -4.8213  <.0001  -3.6292  -1.5313  ***
countryCanada        -1.0578  0.3374  -3.1347  0.0017  -1.7192  -0.3964   **
countryChina         -0.9072  0.3375  -2.6883  0.0072  -1.5686  -0.2458   **
countryItaly         -0.1135  0.8191  -0.1385  0.8898  -1.7189   1.4919
countryNew Zealand   -3.0493  1.4132  -2.1577  0.0310  -5.8191  -0.2794    *
countryUK            -1.3068  0.7087  -1.8441  0.0652  -2.6958   0.0821    .
countryUSA           -0.7114  0.3616  -1.9671  0.0492  -1.4202  -0.0026    *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_country$comp
```

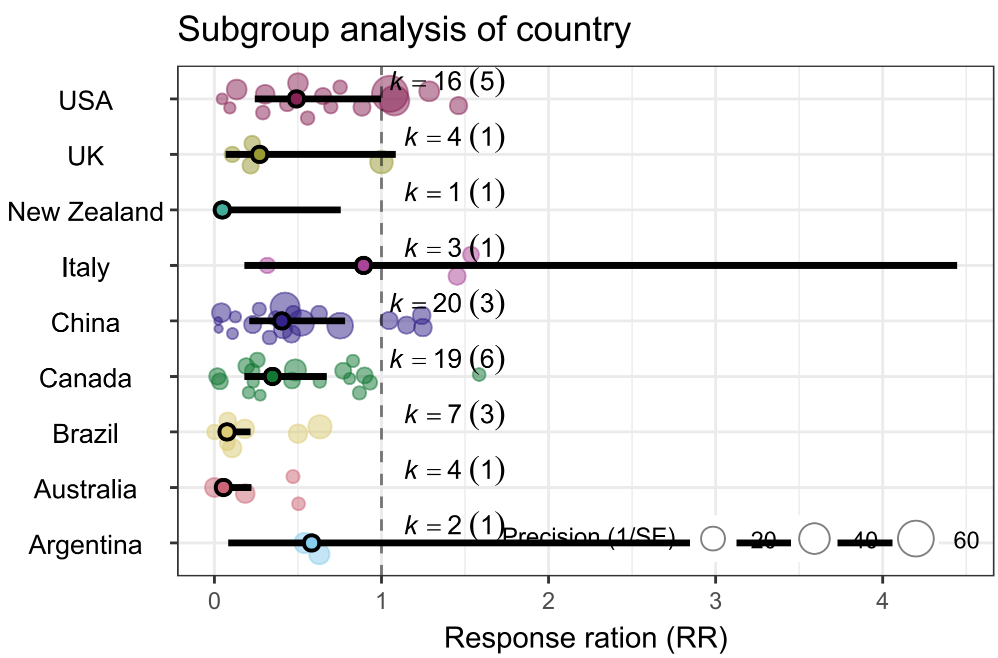
```
# A tibble: 36 × 6
   level1    level2        diff    se      z      p
   <chr>     <chr>        <dbl> <dbl>  <dbl>  <dbl>
 1 Argentina Australia    2.38  1.23   1.93  0.0533
 2 Argentina Brazil       2.04  1.13   1.80  0.0715
 3 Argentina Canada       0.516 1.05   0.491 0.624
 4 Argentina China        0.366 1.05   0.348 0.728
 5 Argentina Italy       -0.428 1.29  -0.332 0.740
 6 Argentina New Zealand  2.51  1.73   1.45  0.147
 7 Argentina UK           0.765 1.22   0.626 0.531
 8 Argentina USA          0.170 1.06   0.160 0.873
 9 Australia Brazil      -0.337 0.897 -0.376 0.707
10 Australia Canada      -1.86  0.795 -2.34  0.0193
# ℹ 26 more rows
```

```
mod_country$mod_res
```

```
         name estimate lowerCL upperCL lowerPR upperPR
1   Argentina    0.582   0.083   4.103   0.020  17.075
2   Australia    0.054   0.013   0.222   0.002   1.198
3      Brazil    0.076   0.027   0.216   0.004   1.448
4      Canada    0.347   0.179   0.673   0.020   5.918
```

```
5       China    0.404   0.208   0.782   0.024   6.880
6       Italy    0.893   0.179   4.446   0.037  21.701
7 New Zealand    0.047   0.003   0.756   0.001   2.361
8          UK    0.271   0.067   1.086   0.012   5.935
9         USA    0.491   0.242   0.997   0.028   8.464
```

```
mod_country$fig
```



7.3 season

```
mod_season <- sub_meta(dat = es, mod = "season", transfm = "exp",
  es.lab="Response ration (RR)", refline.pos = 1)

mod_season$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     1.8669 (SE = 0.3320)
tau (square root of estimated tau^2 value):             1.3664
I^2 (residual heterogeneity / unaccounted variability): 99.55%
H^2 (unaccounted variability / sampling variability):   222.74

Test for Residual Heterogeneity:
```

```
QE(df = 72) = 7460.2468, p-val < .0001

Test of Moderators (coefficients 1:4):
QM(df = 4) = 72.1487, p-val < .0001

Model Results:

              estimate      se     zval    pval    ci.lb    ci.ub
seasonautumn   -1.4985  0.4637  -3.2318  0.0012  -2.4073  -0.5897   **
seasonspring   -0.9778  0.2764  -3.5377  0.0004  -1.5195  -0.4361  ***
seasonsummer   -0.7139  0.2561  -2.7881  0.0053  -1.2158  -0.2121   **
seasonwinter   -2.9778  0.4627  -6.4355  <.0001  -3.8847  -2.0709  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

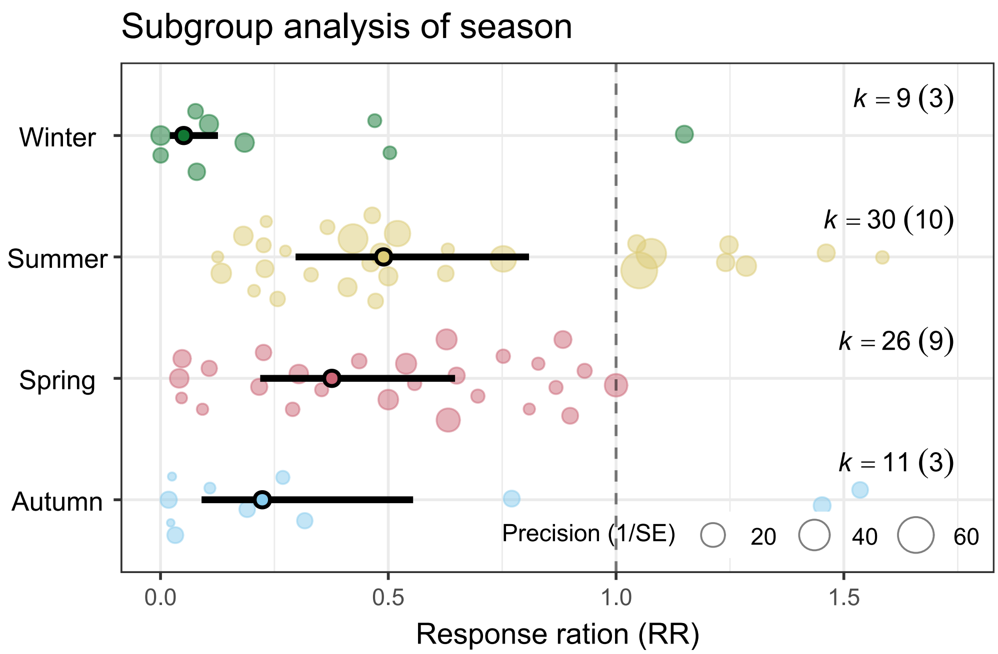
```
mod_season$comp
```

```
# A tibble: 6 × 6
  level1 level2   diff    se      z         p
  <chr>  <chr>   <dbl> <dbl>  <dbl>     <dbl>
1 autumn spring -0.521 0.540 -0.965 0.335
2 autumn summer -0.785 0.530 -1.48  0.139
3 autumn winter  1.48  0.655  2.26  0.0239
4 spring summer -0.264 0.377 -0.700 0.484
5 spring winter  2.00  0.539  3.71  0.000207
6 summer winter  2.26  0.529  4.28  0.0000186
```

```
mod_season$mod_res
```

```
    name estimate lowerCL upperCL lowerPR upperPR
1 Autumn    0.223   0.090   0.554   0.013   3.779
2 Spring    0.376   0.219   0.647   0.024   5.781
3 Summer    0.490   0.296   0.809   0.032   7.469
4 Winter    0.051   0.021   0.126   0.003   0.860
```

```
mod_season$fig
```



7.4 Measurement method

```
mod_mea <- sub_meta(dat = es, mod = "mea_method",
                         transfm = "exp", es.lab="Response ration (RR)")

mod_mea$model
```

```
Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.3925 (SE = 0.4214)
tau (square root of estimated tau^2 value):             1.5468
I^2 (residual heterogeneity / unaccounted variability): 99.65%
H^2 (unaccounted variability / sampling variability):   285.22

Test for Residual Heterogeneity:
QE(df = 72) = 9806.9914, p-val < .0001

Test of Moderators (coefficients 1:4):
QM(df = 4) = 42.6735, p-val < .0001

Model Results:

               estimate      se     zval    pval    ci.lb    ci.ub
mea_methodDC    -1.4145  0.3818  -3.7046  0.0002  -2.1629  -0.6661  ***
mea_methodIHF   -1.2765  0.5037  -2.5341  0.0113  -2.2638  -0.2892    *
```

```
mea_methodSC     -1.2078  0.3398  -3.5544  0.0004  -1.8738  -0.5418  ***
mea_methodWT     -0.9823  0.3123  -3.1455  0.0017  -1.5944  -0.3702   **

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

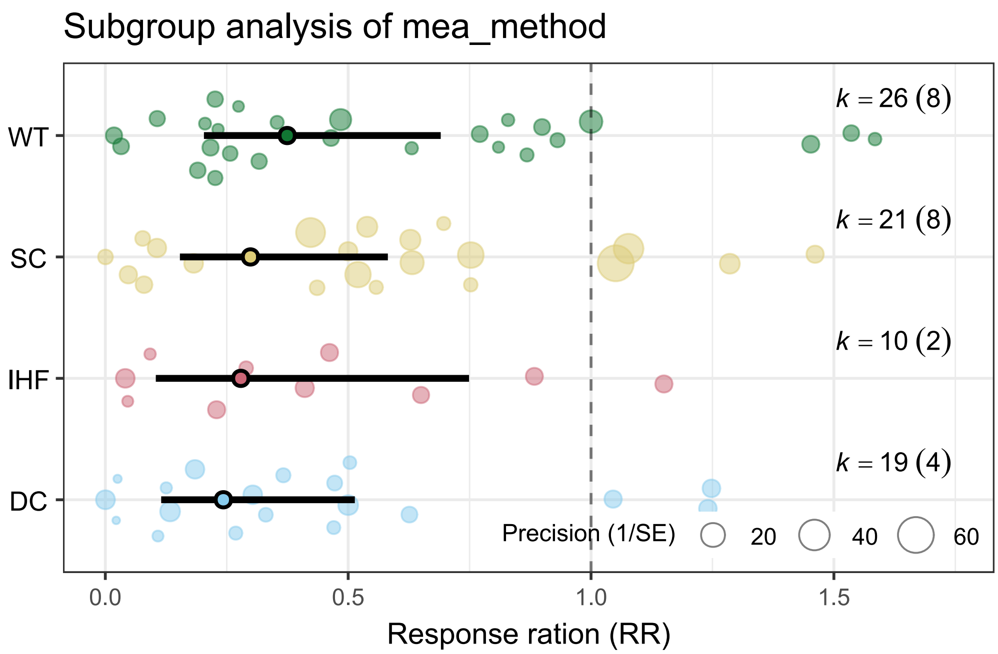
```
mod_mea$comp
```

```
# A tibble: 6 × 6
  level1 level2    diff    se      z     p
  <chr>  <chr>    <dbl> <dbl>  <dbl> <dbl>
1 DC     IHF    -0.138  0.632 -0.218 0.827
2 DC     SC     -0.207  0.511 -0.404 0.686
3 DC     WT     -0.432  0.493 -0.876 0.381
4 IHF    SC     -0.0687 0.608 -0.113 0.910
5 IHF    WT     -0.294  0.593 -0.496 0.620
6 SC     WT     -0.225  0.462 -0.489 0.625
```

```
mod_mea$mod_res
```

```
  name estimate lowerCL upperCL lowerPR upperPR
1   DC    0.243   0.115   0.514   0.011   5.519
2  IHF    0.279   0.104   0.749   0.012   6.765
3   SC    0.299   0.154   0.582   0.013   6.660
4   WT    0.374   0.203   0.691   0.017   8.252
```

```
mod_mea$fig
```



7.5 N type

```
mod_Ntype <- sub_meta(dat = es, mod = "N_type", transfm = "exp",
  es.lab="Response ration (RR)", refline.pos = 1)

mod_Ntype$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     1.6722 (SE = 0.3071)
tau (square root of estimated tau^2 value):             1.2931
I^2 (residual heterogeneity / unaccounted variability): 99.54%
H^2 (unaccounted variability / sampling variability):   217.22

Test for Residual Heterogeneity:
QE(df = 68) = 5967.1561, p-val < .0001

Test of Moderators (coefficients 1:8):
QM(df = 8) = 91.3469, p-val < .0001

Model Results:

                 estimate      se     zval    pval     ci.lb    ci.ub
N_typeAB          -0.6337  0.6597  -0.9605  0.3368   -1.9266   0.6593
N_typeAN          -8.1887  1.2972  -6.3125  <.0001  -10.7312  -5.6462  ***
```

```
N_typeAS           -0.3615  1.3466  -0.2685  0.7884   -3.0008   2.2778
N_typeASN          -0.2846  1.3397  -0.2125  0.8317   -2.9103   2.3411
N_typeUAN          -0.9676  0.4345  -2.2270  0.0260   -1.8191  -0.1160    *
N_typeUrea         -1.2277  0.1830  -6.7078  <.0001   -1.5864  -0.8689  ***
N_typeUrea+UI      -0.6404  0.9332  -0.6862  0.4926   -2.4695   1.1887
N_typeUrea+UINI    -0.1506  0.8111  -0.1857  0.8527   -1.7403   1.4391

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

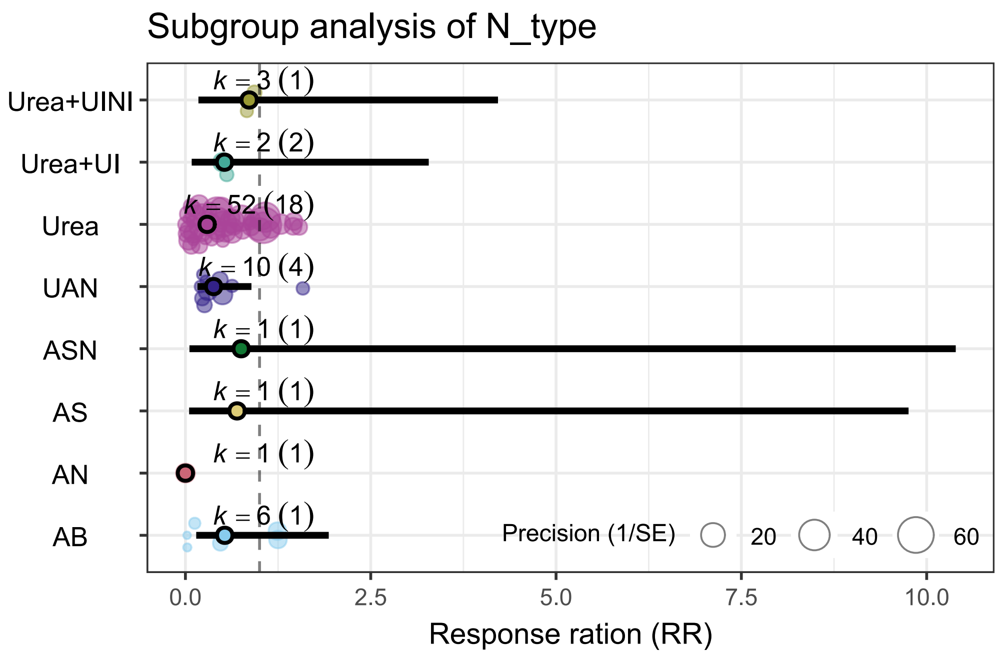
```
mod_Ntype$comp
```

```
# A tibble: 28 × 6
   level1 level2        diff    se         z           p
   <chr>  <chr>        <dbl> <dbl>     <dbl>       <dbl>
 1 AB     AN         7.56    1.46   5.19    0.000000209
 2 AB     AS        -0.272   1.50  -0.181   0.856
 3 AB     ASN       -0.349   1.49  -0.234   0.815
 4 AB     UAN        0.334   0.790  0.423   0.673
 5 AB     Urea       0.594   0.685  0.868   0.386
 6 AB     Urea+UI    0.00674 1.14   0.00590 0.995
 7 AB     Urea+UINI -0.483   1.05  -0.462   0.644
 8 AN     AS        -7.83    1.87  -4.19    0.0000284
 9 AN     ASN       -7.90    1.86  -4.24    0.0000225
10 AN     UAN       -7.22    1.37  -5.28    0.000000130
# ℹ 18 more rows
```

```
mod_Ntype$mod_res
```

```
       name estimate lowerCL upperCL lowerPR upperPR
1        AB    0.531   0.146   1.933   0.031   9.130
2        AN    0.000   0.000   0.004   0.000   0.010
3        AS    0.697   0.050   9.756   0.018  27.051
4       ASN    0.752   0.054  10.392   0.020  28.926
5       UAN    0.380   0.162   0.890   0.026   5.508
6      Urea    0.293   0.205   0.419   0.023   3.789
7   Urea+UI    0.527   0.085   3.283   0.023  12.003
8 Urea+UINI    0.860   0.175   4.217   0.043  17.137
```

```
mod_Ntype$fig
```



7.6 Nrate

```
mod_Nrate <- sub_meta(dat = es, mod = "rate", transfm = "none",
  es.lab="ln(RR)")

mod_Nrate$model
```

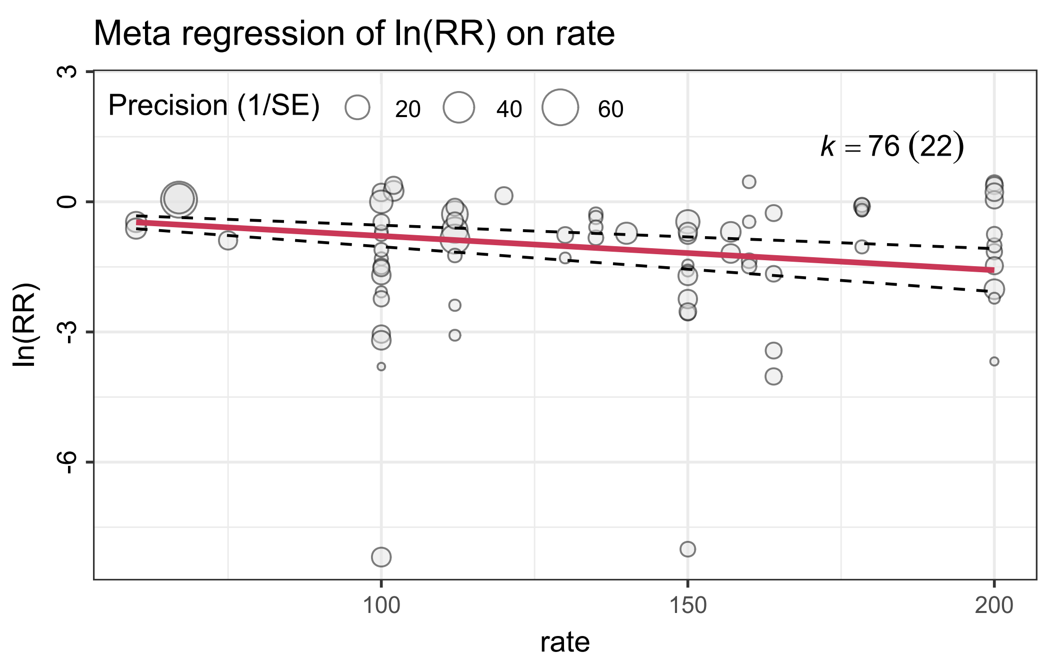
```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.3122 (SE = 0.3992)
tau (square root of estimated tau^2 value):             1.5206
I^2 (residual heterogeneity / unaccounted variability): 99.68%
H^2 (unaccounted variability / sampling variability):   309.63

Test for Residual Heterogeneity:
QE(df = 75) = 14673.1602, p-val < .0001

Model Results:

      estimate      se     zval    pval    ci.lb    ci.ub
rate   -0.0079  0.0013  -6.2264  <.0001  -0.0104  -0.0054  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_Nrate$comp
```

```
Parameter1 | Parameter2 |    r |         95% CI | t(74) |     p
----------------------------------------------------------------
rate       |         yi | 0.03 | [-0.19, 0.26] |  0.29 | 0.769

Observations: 76
```

```
mod_Nrate$fig
```

```
Ignoring unknown labels:
• parse : "TRUE"
```



7.7 Application method

```
mod_sub.tech <- sub_meta(dat = es, mod = "sub_tech", transfm = "exp",
  es.lab="Response ration (RR)")

mod_sub.tech$model
```

```
Mixed-Effects Model (k = 76; tau^2 estimator: REML)
```

```
tau^2 (estimated amount of residual heterogeneity):     2.0119 (SE = 0.3566)
tau (square root of estimated tau^2 value):             1.4184
I^2 (residual heterogeneity / unaccounted variability): 99.50%
H^2 (unaccounted variability / sampling variability):   201.78

Test for Residual Heterogeneity:
QE(df = 72) = 8968.6983, p-val < .0001

Test of Moderators (coefficients 1:4):
QM(df = 4) = 62.1271, p-val < .0001

Model Results:

                    estimate      se     zval    pval    ci.lb    ci.ub
sub_techincorp       -1.8177  0.2678  -6.7880  <.0001  -2.3425  -1.2929  ***
sub_techinjection    -0.6144  0.3934  -1.5619  0.1183  -1.3855   0.1566
sub_techIrrigation   -1.4050  0.4043  -3.4749  0.0005  -2.1975  -0.6125  ***
sub_techmix          -0.4239  0.3420  -1.2395  0.2152  -1.0943   0.2464

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

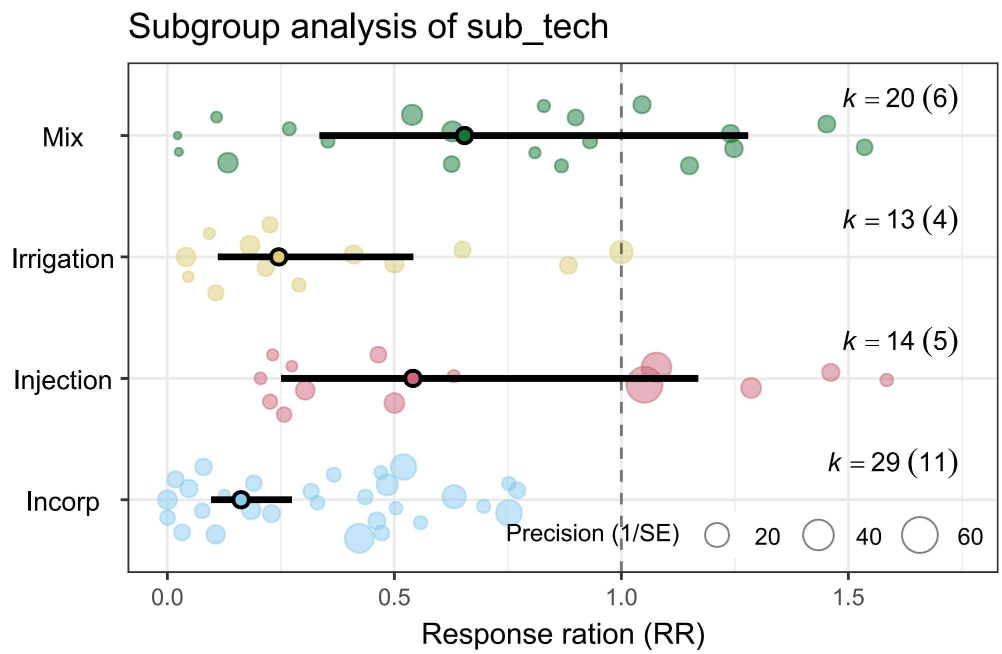
```
mod_sub.tech$comp
```

```
# A tibble: 6 × 6
  level1     level2       diff    se      z       p
  <chr>      <chr>       <dbl> <dbl>  <dbl>   <dbl>
1 incorp     injection  -1.20  0.476 -2.53  0.0115
2 incorp     Irrigation -0.413 0.485 -0.851 0.395
3 incorp     mix        -1.39  0.434 -3.21  0.00133
4 injection  Irrigation  0.791 0.564  1.40  0.161
5 injection  mix        -0.191 0.521 -0.365 0.715
6 Irrigation mix        -0.981 0.530 -1.85  0.0640
```

```
mod_sub.tech$mod_res
```

```
        name estimate lowerCL upperCL lowerPR upperPR
1     Incorp    0.162   0.096   0.274   0.010   2.750
2  Injection    0.541   0.250   1.170   0.030   9.685
3 Irrigation    0.245   0.111   0.542   0.014   4.418
4        Mix    0.654   0.335   1.279   0.037  11.425
```

```
mod_sub.tech$fig
```



7.8 Fully vs Partly

```
mod_sub.type <- es |>
    filter(sub_tech!="Irrigation") |>
    sub_meta(mod = "sub_type",  transfm = "exp", es.lab="Response ration (RR)")

mod_sub.type$model
```

```


Mixed-Effects Model (k = 63; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.1935 (SE = 0.4218)
tau (square root of estimated tau^2 value):             1.4810
I^2 (residual heterogeneity / unaccounted variability): 99.60%
H^2 (unaccounted variability / sampling variability):   251.87

Test for Residual Heterogeneity:
QE(df = 61) = 8262.1863, p-val < .0001

Test of Moderators (coefficients 1:2):
QM(df = 2) = 45.9302, p-val < .0001

Model Results:

                estimate      se     zval    pval    ci.lb    ci.ub
sub_typefully    -1.8170  0.2793  -6.5066  <.0001  -2.3643  -1.2697  ***
```

```
sub_typepartly   -0.5097  0.2688  -1.8959  0.0580  -1.0366   0.0172    .

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

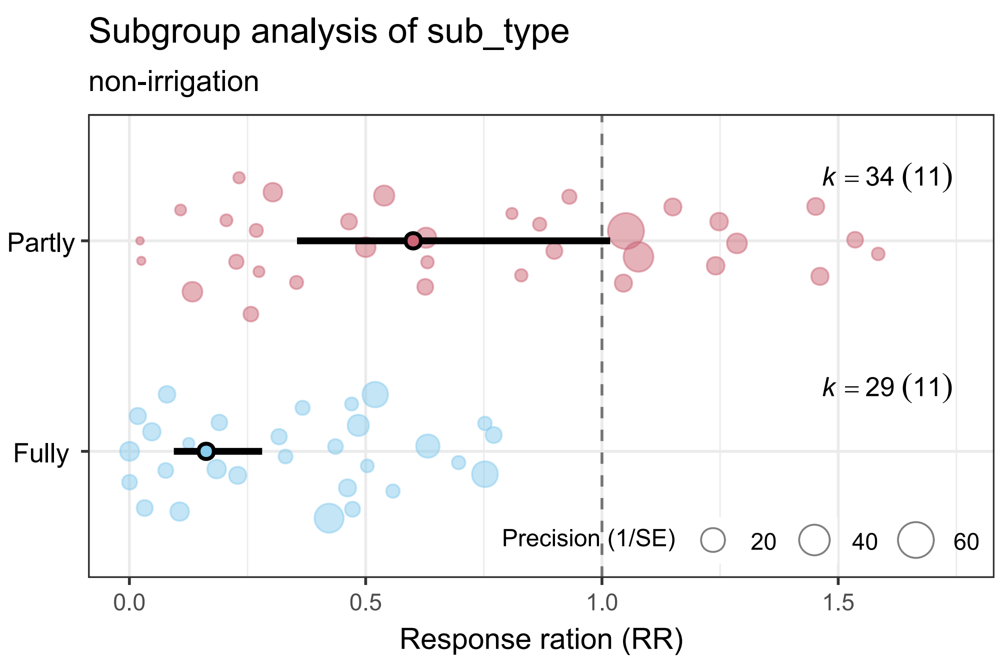
```
mod_sub.type$comp
```

```
# A tibble: 1 × 6
  level1 level2  diff    se     z         p
  <chr>  <chr>  <dbl> <dbl> <dbl>     <dbl>
1 fully  partly -1.31 0.388 -3.37 0.000744
```

```
mod_sub.type$mod_res
```

```
    name estimate lowerCL upperCL lowerPR upperPR
1  Fully    0.163   0.094   0.281   0.008   3.117
2 Partly    0.601   0.355   1.017   0.031  11.479
```

```
mod_sub.type$fig+
  labs(subtitle="non-irrigation")
```



7.9 Incorporation/injection depth

```
mod_sub.depth <- sub_meta(dat = es[es$sub_tech!="Irrigation",], mod =
"sub_depth",
                         transfm = "none", es.lab="ln(RR)")

mod_sub.depth$model
```

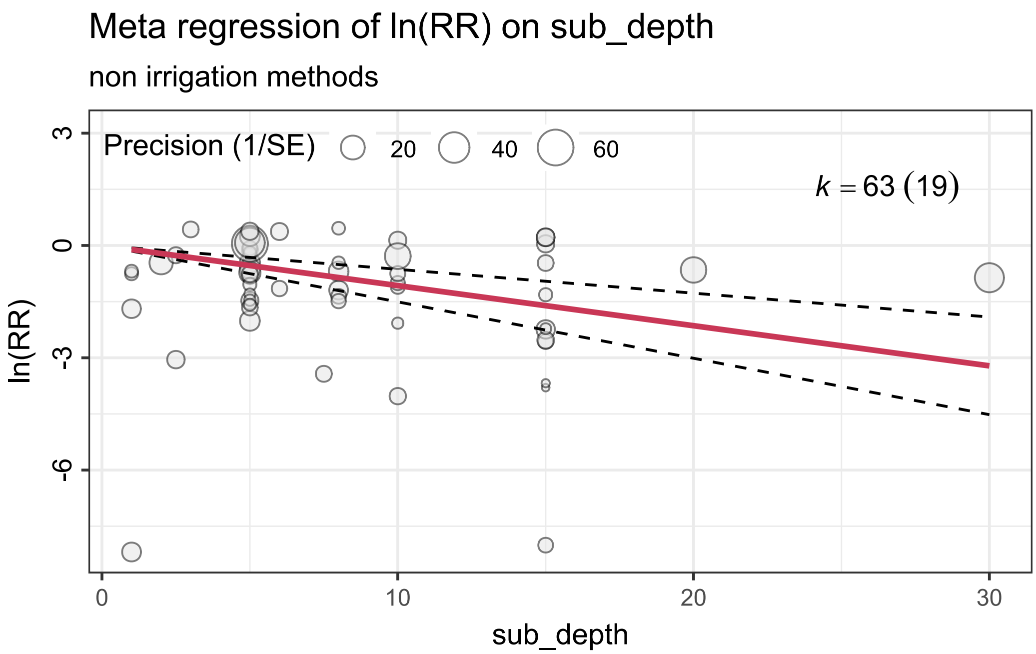
```

Mixed-Effects Model (k = 63; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.6121 (SE = 0.4951)
tau (square root of estimated tau^2 value):             1.6162
I^2 (residual heterogeneity / unaccounted variability): 99.74%
H^2 (unaccounted variability / sampling variability):   378.59

Test for Residual Heterogeneity:
QE(df = 62) = 14828.4955, p-val < .0001

Model Results:

           estimate      se     zval    pval    ci.lb    ci.ub
sub_depth   -0.1072  0.0222  -4.8320  <.0001  -0.1507  -0.0637  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_sub.depth$comp
```

```
Parameter1 | Parameter2 |     r |        95% CI | t(61) |     p
---------------------------------------------------------------
sub_depth  |         yi | -0.14 | [-0.38, 0.11] | -1.11 | 0.273

Observations: 63
```

```
mod_sub.depth$fig+
    labs(subtitle="non irrigation methods")
```

```
Ignoring unknown labels:
• parse : "TRUE"
```




```r
mod_sub.depth2 <- sub_meta(dat = es[es$sub_tech!="Irrigation",], mod =
"sub_depth2",
                         transfm = "exp", es.lab="Response ration (RR)")
mod_sub.depth2$model
```

```
Mixed-Effects Model (k = 63; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.6150 (SE = 0.5041)
tau (square root of estimated tau^2 value):             1.6171
I^2 (residual heterogeneity / unaccounted variability): 99.65%
H^2 (unaccounted variability / sampling variability):   285.68

Test for Residual Heterogeneity:
QE(df = 60) = 10029.5655, p-val < .0001

Test of Moderators (coefficients 1:3):
QM(df = 3) = 31.3719, p-val < .0001

Model Results:

                estimate      se     zval    pval    ci.lb    ci.ub
sub_depth2<5     -0.9287  0.2877  -3.2283  0.0012  -1.4926  -0.3649   **
sub_depth25~10   -1.1243  0.4116  -2.7316  0.0063  -1.9309  -0.3176   **
sub_depth2>10    -1.7239  0.4694  -3.6727  0.0002  -2.6439  -0.8039  ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

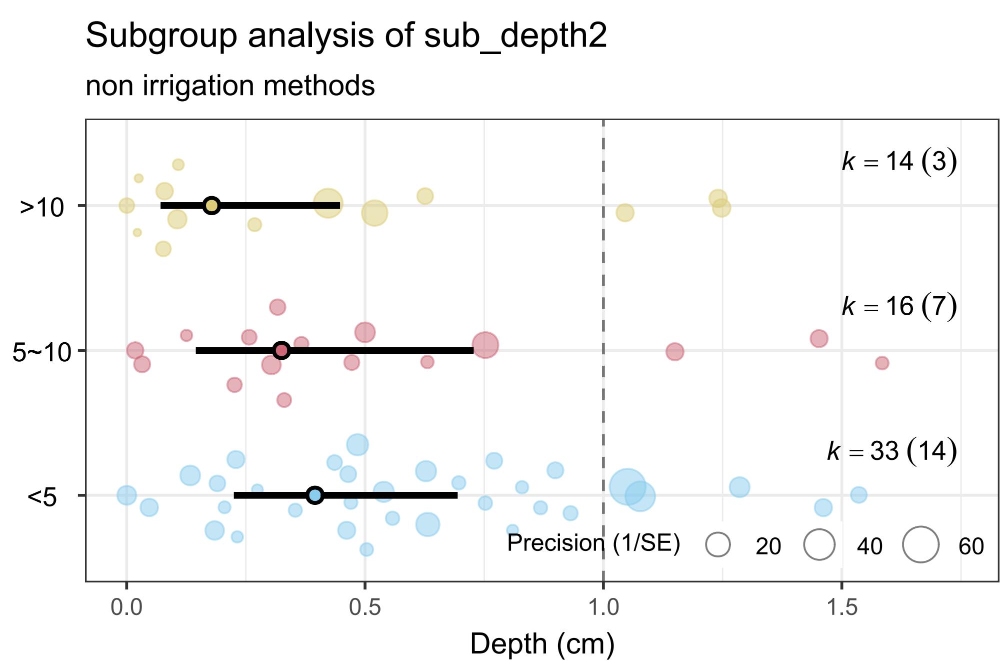
```
mod_sub.depth2$comp
```

```
# A tibble: 3 × 6
  level1 level2  diff    se     z     p
  <chr>  <chr>  <dbl> <dbl> <dbl> <dbl>
1 <5     5~10   0.196 0.502 0.389 0.697
2 <5     >10    0.795 0.551 1.44  0.149
3 5~10   >10    0.600 0.624 0.961 0.337
```

```
mod_sub.depth2$mod_res
```

```
  name estimate lowerCL upperCL lowerPR upperPR
1   <5    0.395   0.225   0.694   0.016   9.880
2 5~10    0.325   0.145   0.728   0.012   8.552
3  >10    0.178   0.071   0.448   0.007   4.837
```

```
mod_sub.depth2$fig+
    labs(subtitle="non irrigation methods", y="Depth (cm)")
```



7.10 Irrigation amount

```
mod_sub.depth <- sub_meta(dat = es[es$sub_tech=="Irrigation",], mod =
"sub_depth",
                         transfm = "none", es.lab="ln(RR)")

mod_sub.depth$model
```

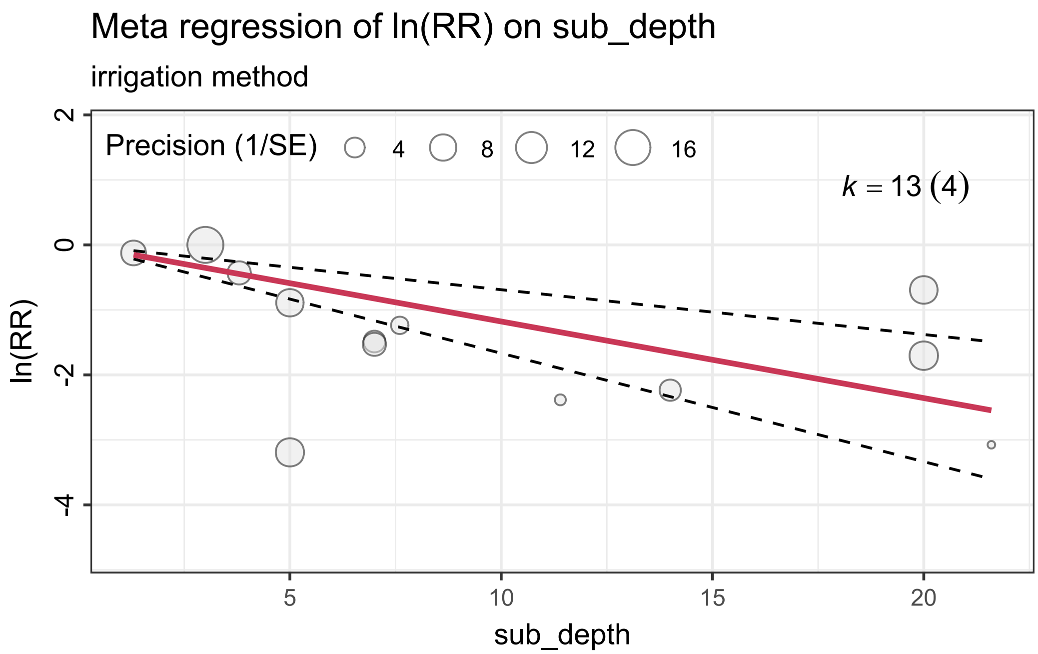
```

Mixed-Effects Model (k = 13; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     0.9456 (SE = 0.4197)
tau (square root of estimated tau^2 value):             0.9724
I^2 (residual heterogeneity / unaccounted variability): 98.27%
H^2 (unaccounted variability / sampling variability):   57.68

Test for Residual Heterogeneity:
QE(df = 12) = 1082.7553, p-val < .0001

Model Results:

           estimate      se      zval    pval    ci.lb    ci.ub
sub_depth   -0.1179  0.0250  -4.7234  <.0001  -0.1668  -0.0689  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_sub.depth$comp
```

```
Parameter1 | Parameter2 |     r |        95% CI | t(11) |     p
---------------------------------------------------------------
sub_depth  |         yi | -0.46 | [-0.81, 0.12] | -1.73 | 0.111

Observations: 13
```

```
mod_sub.depth$fig+
    labs(subtitle="irrigation method")
```

```
Ignoring unknown labels:
• parse : "TRUE"
```




```r
mod_sub.depth2_irr <- sub_meta(dat = es[es$sub_tech!="Irrigation",], mod =
"sub_depth2",
                          transfm = "exp", es.lab="Response ration (RR)")

mod_sub.depth2$model
```

```

Mixed-Effects Model (k = 63; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.6150 (SE = 0.5041)
tau (square root of estimated tau^2 value):             1.6171
I^2 (residual heterogeneity / unaccounted variability): 99.65%
H^2 (unaccounted variability / sampling variability):   285.68

Test for Residual Heterogeneity:
QE(df = 60) = 10029.5655, p-val < .0001

Test of Moderators (coefficients 1:3):
QM(df = 3) = 31.3719, p-val < .0001

Model Results:

                estimate      se     zval    pval    ci.lb    ci.ub
sub_depth2<5     -0.9287  0.2877  -3.2283  0.0012  -1.4926  -0.3649   **
sub_depth25~10   -1.1243  0.4116  -2.7316  0.0063  -1.9309  -0.3176   **
sub_depth2>10    -1.7239  0.4694  -3.6727  0.0002  -2.6439  -0.8039  ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

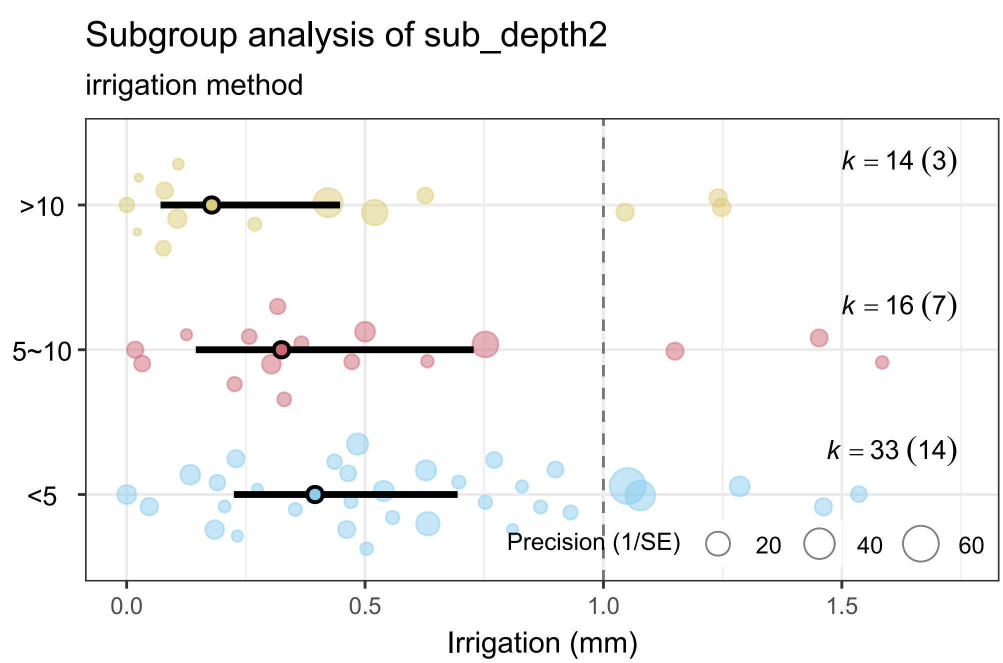
```r
mod_sub.depth2$comp
```

```
# A tibble: 3 × 6
  level1 level2  diff    se     z     p
  <chr>  <chr>  <dbl> <dbl> <dbl> <dbl>
1 <5     5~10   0.196 0.502 0.389 0.697
2 <5     >10    0.795 0.551 1.44  0.149
3 5~10   >10    0.600 0.624 0.961 0.337
```

```r
mod_sub.depth2_irr$mod_res
```

```
  name estimate lowerCL upperCL lowerPR upperPR
1   <5    0.395   0.225   0.694   0.016   9.880
2 5~10    0.325   0.145   0.728   0.012   8.552
3  >10    0.178   0.071   0.448   0.007   4.837
```

```r
mod_sub.depth2_irr$fig+
    labs(subtitle="irrigation method", y="Irrigation (mm)")
```



7.11 Soil texture

```r
mod_soil.text2 <- sub_meta(dat = es, mod = "soil_texture2",
                         transfm = "exp", es.lab="Response ration (RR)")

mod_soil.text2$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2211 (SE = 0.3894)
tau (square root of estimated tau^2 value):             1.4903
I^2 (residual heterogeneity / unaccounted variability): 99.56%
H^2 (unaccounted variability / sampling variability):   225.97

Test for Residual Heterogeneity:
QE(df = 73) = 9370.5314, p-val < .0001

Test of Moderators (coefficients 1:3):
QM(df = 3) = 49.2234, p-val < .0001

Model Results:

                    estimate      se     zval    pval    ci.lb    ci.ub
soil_texture2clay    -1.7447  0.3210  -5.4352  <.0001  -2.3738  -1.1155  ***
soil_texture2loam    -0.9507  0.2521  -3.7706  0.0002  -1.4449  -0.4565  ***
soil_texture2sandy   -0.9188  0.3930  -2.3377  0.0194  -1.6892  -0.1485    *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
mod_soil.text2$comp
```

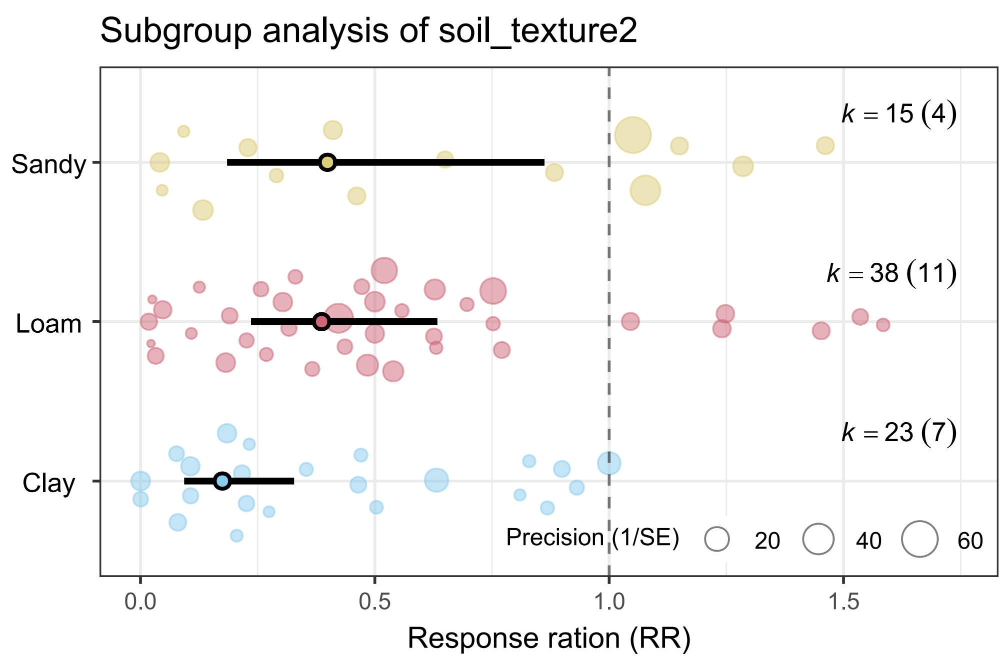
```
# A tibble: 3 × 6
  level1 level2    diff    se       z      p
  <chr>  <chr>    <dbl> <dbl>   <dbl>  <dbl>
1 clay   loam   -0.794  0.408 -1.95   0.0518
2 clay   sandy  -0.826  0.507 -1.63   0.104
3 loam   sandy  -0.0319 0.467 -0.0684 0.945
```

```r
mod_soil.text2$mod_res
```

```
   name estimate lowerCL upperCL lowerPR upperPR
1  Clay    0.175   0.093   0.328   0.009   3.467
```

```
2  Loam     0.386   0.236   0.633   0.020   7.477
3 Sandy     0.399   0.185   0.862   0.019   8.183
```

```
mod_soil.text2$fig
```



7.12 Soil caly

```
mod_soil.clay <- sub_meta(dat = es, mod = "soil_clay",
                          transfm = "none", es.lab="ln(RR)")
mod_soil.clay$model
```

```


Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.3212 (SE = 0.4006)
tau (square root of estimated tau^2 value):             1.5235
I^2 (residual heterogeneity / unaccounted variability): 99.68%
H^2 (unaccounted variability / sampling variability):   308.65

Test for Residual Heterogeneity:
QE(df = 75) = 14954.3807, p-val < .0001

Model Results:

```

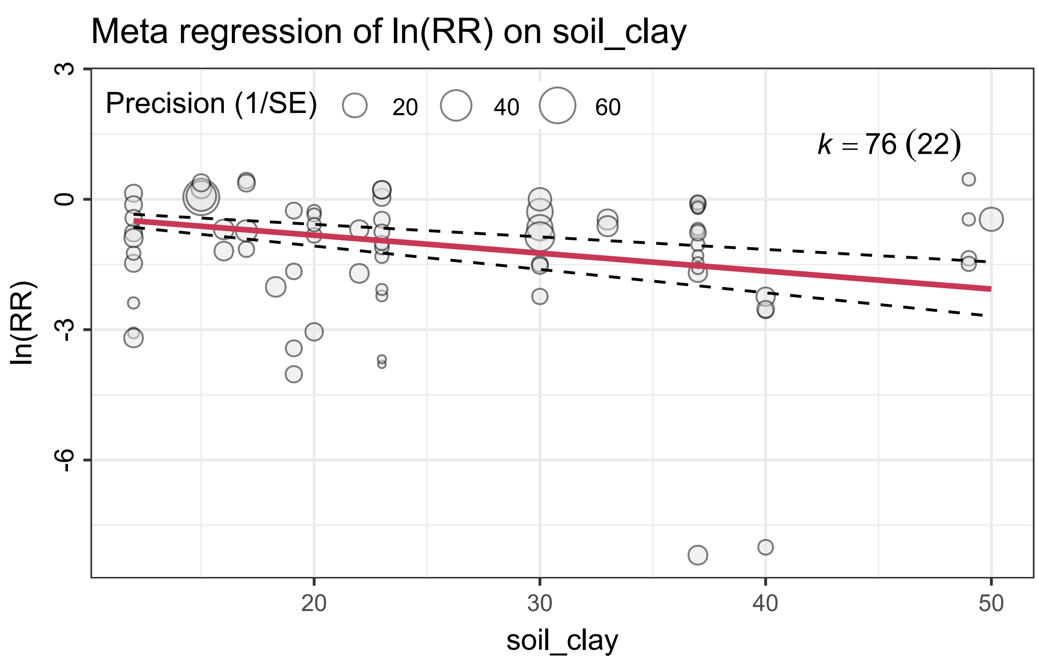
```
           estimate      se      zval     pval     ci.lb     ci.ub
soil_clay   -0.0413  0.0064  -6.4893  <.0001  -0.0538  -0.0288  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_soil.clay$comp
```

```
Parameter1 | Parameter2 |     r |        95% CI | t(74) |     p
---------------------------------------------------------------
soil_clay  |         yi | -0.10 | [-0.32, 0.13] | -0.86 | 0.393

Observations: 76
```

```
mod_soil.clay$fig
```

```
Ignoring unknown labels:
• parse : "TRUE"
```




```
mod_soil.clay2 <- sub_meta(dat = es, mod = "soil_clay2",
                         transfm = "exp", es.lab="Response ration (RR)")
```

```
mod_soil.clay2$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2364 (SE = 0.3920)
tau (square root of estimated tau^2 value):             1.4955
I^2 (residual heterogeneity / unaccounted variability): 99.57%
H^2 (unaccounted variability / sampling variability):   232.62

Test for Residual Heterogeneity:
QE(df = 73) = 8664.8594, p-val < .0001

Test of Moderators (coefficients 1:3):
QM(df = 3) = 48.4125, p-val < .0001

Model Results:

                   estimate      se     zval    pval    ci.lb    ci.ub     
soil_clay2<18%      -0.7319  0.3401  -2.1517  0.0314  -1.3985  -0.0652    * 
soil_clay218%~35%   -1.1497  0.2722  -4.2240  <.0001  -1.6831  -0.6162  *** 
soil_clay2>35%      -1.6476  0.3235  -5.0931  <.0001  -2.2816  -1.0135  *** 

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

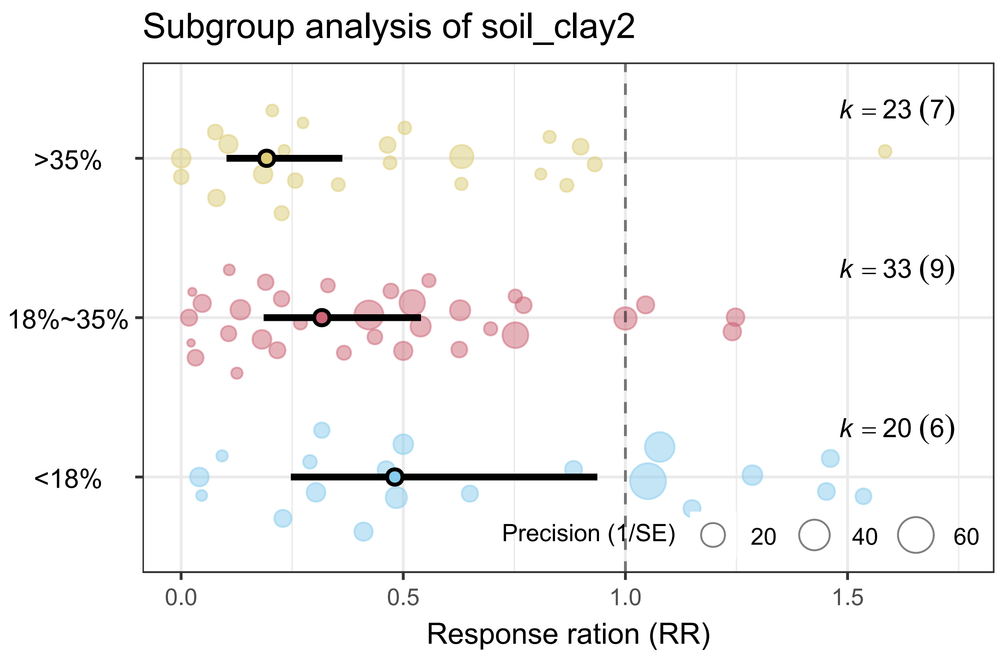
```
mod_soil.clay2$comp
```

```
# A tibble: 3 × 6
  level1  level2   diff    se     z      p
  <chr>   <chr>   <dbl> <dbl> <dbl>  <dbl>
1 <18%    18%~35% 0.418 0.436 0.959 0.338 
2 <18%    >35%    0.916 0.469 1.95  0.0511
3 18%~35% >35%    0.498 0.423 1.18  0.239 
```

```
mod_soil.clay2$mod_res
```

```
     name estimate lowerCL upperCL lowerPR upperPR
1    <18%    0.481   0.247   0.937   0.024   9.718
2 18%~35%    0.317   0.186   0.540   0.016   6.231
3    >35%    0.193   0.102   0.363   0.010   3.862
```

```
mod_soil.clay2$fig
```



7.13 Soil pH

```
mod_soil.pH <- sub_meta(dat = es, mod = "soil_pH",
                        transfm = "none", es.lab="ln(RR)")
mod_soil.pH$model
```

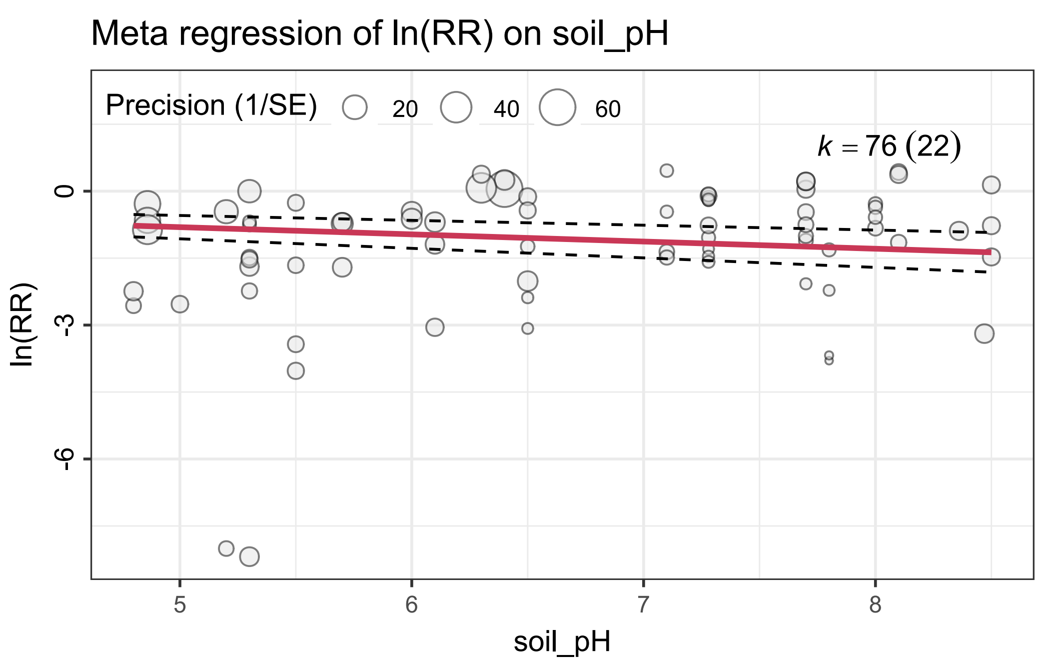
```
Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2935 (SE = 0.3960)
tau (square root of estimated tau^2 value):             1.5144
I^2 (residual heterogeneity / unaccounted variability): 99.64%
H^2 (unaccounted variability / sampling variability):   280.67

Test for Residual Heterogeneity:
QE(df = 75) = 11386.3760, p-val < .0001

Model Results:

         estimate      se     zval    pval    ci.lb    ci.ub
soil_pH   -0.1612  0.0267  -6.0441  <.0001  -0.2134  -0.1089  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
mod_soil.pH$comp
```

```
Parameter1 | Parameter2 |    r |       95% CI | t(74) |      p
---------------------------------------------------------------
soil_pH    |         yi | 0.26 | [0.04, 0.46] |  2.35 | 0.022*

Observations: 76
```

```r
mod_soil.pH$fig
```

```
Ignoring unknown labels:
• parse : "TRUE"
```




```r
mod_soil.pH2 <- sub_meta(dat = es, mod = "soil_pH2",
                         transfm = "exp", es.lab="Response ration (RR)")
mod_soil.pH2$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.0998 (SE = 0.3717)
tau (square root of estimated tau^2 value):             1.4491
```

```
I^2 (residual heterogeneity / unaccounted variability): 99.54%
H^2 (unaccounted variability / sampling variability):   219.38

Test for Residual Heterogeneity:
QE(df = 72) = 9592.7682, p-val < .0001

Test of Moderators (coefficients 1:4):
QM(df = 4) = 56.9532, p-val < .0001

Model Results:

             estimate      se     zval    pval    ci.lb    ci.ub
soil_pH2<6    -1.9091  0.2921  -6.5367  <.0001  -2.4815  -1.3366  ***
soil_pH26~7   -0.9618  0.4120  -2.3347  0.0196  -1.7693  -0.1544    *
soil_pH27~8   -0.7131  0.2853  -2.4991  0.0125  -1.2723  -0.1538    *
soil_pH2>8    -0.8191  0.5151  -1.5902  0.1118  -1.8287   0.1905

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

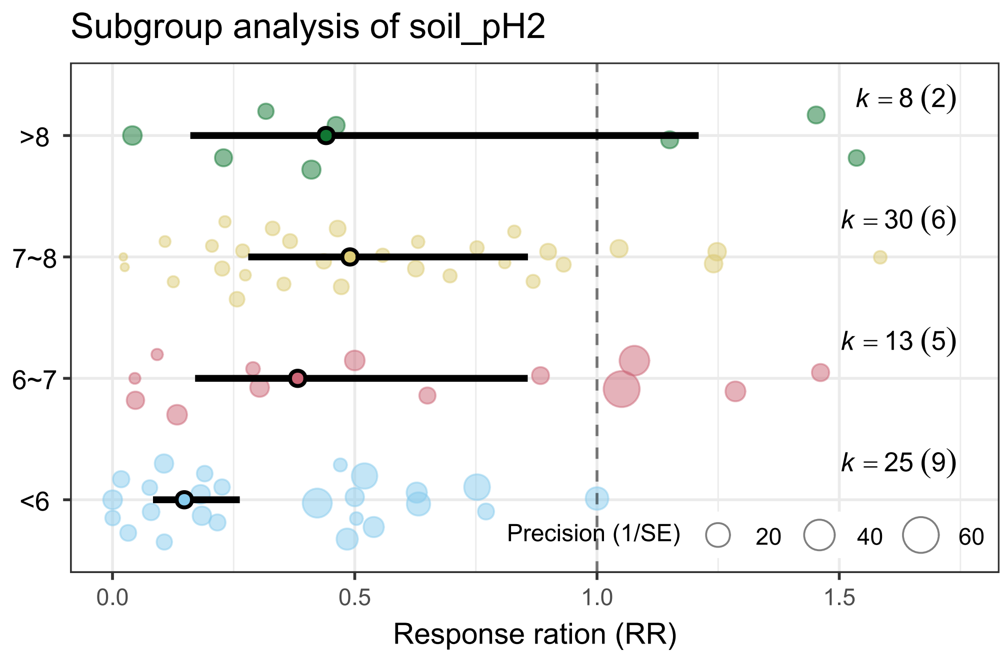
```
mod_soil.pH2$comp
```

```
# A tibble: 6 × 6
  level1 level2   diff    se      z       p
  <chr>  <chr>   <dbl> <dbl>  <dbl>   <dbl>
1 <6     6~7    -0.947 0.505 -1.88  0.0607
2 <6     7~8    -1.20  0.408 -2.93  0.00340
3 <6     >8     -1.09  0.592 -1.84  0.0657
4 6~7    7~8    -0.249 0.501 -0.496 0.620
5 6~7    >8     -0.143 0.660 -0.216 0.829
6 7~8    >8      0.106 0.589  0.180 0.857
```

```
mod_soil.pH2$mod_res
```

```
  name estimate lowerCL upperCL lowerPR upperPR
1   <6    0.148   0.084   0.263   0.008   2.686
2  6~7    0.382   0.170   0.857   0.020   7.322
3  7~8    0.490   0.280   0.857   0.027   8.861
4   >8    0.441   0.161   1.210   0.022   8.981
```

```
mod_soil.pH2$fig
```



7.14 SOC

```
mod_SOC <- sub_meta(dat = es, mod = "SOC",
                         transfm = "none", es.lab="ln(RR)")

mod_SOC$model
```

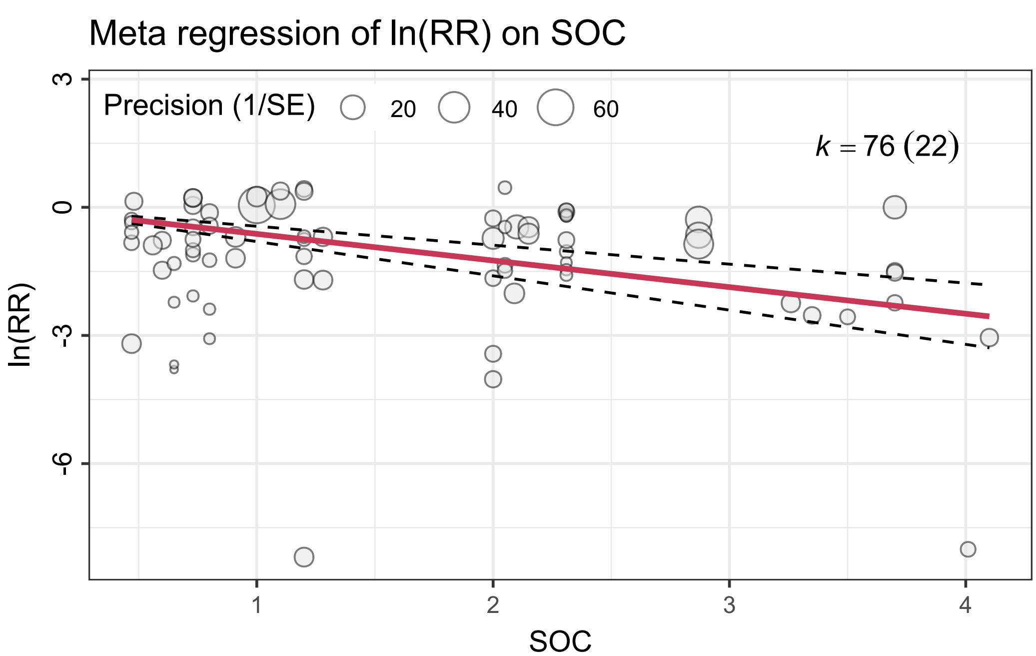
```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2987 (SE = 0.3970)
tau (square root of estimated tau^2 value):             1.5162
I^2 (residual heterogeneity / unaccounted variability): 99.67%
H^2 (unaccounted variability / sampling variability):   307.49

Test for Residual Heterogeneity:
QE(df = 75) = 15221.6246, p-val < .0001

Model Results:

     estimate      se      zval    pval    ci.lb    ci.ub
SOC   -0.6228  0.0915  -6.8096  <.0001  -0.8021  -0.4436  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_SOC$comp
```

```
Parameter1 | Parameter2 |     r |        95% CI | t(74) |     p
----------------------------------------------------------------
SOC        |         yi | -0.21 | [-0.41, 0.02] | -1.80 | 0.075

Observations: 76
```

```
mod_SOC$fig
```

```
Ignoring unknown labels:
• parse : "TRUE"
```




```
mod_SOC2 <- sub_meta(dat = es, mod = "SOC2",
                     transfm = "exp", es.lab="Response ration (RR)")
mod_SOC2$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.3028 (SE = 0.4033)
tau (square root of estimated tau^2 value):             1.5175
```

```
I^2 (residual heterogeneity / unaccounted variability): 99.51%
H^2 (unaccounted variability / sampling variability):   204.92

Test for Residual Heterogeneity:
QE(df = 73) = 10725.4417, p-val < .0001

Test of Moderators (coefficients 1:3):
QM(df = 3) = 45.2386, p-val < .0001

Model Results:

           estimate      se     zval    pval    ci.lb    ci.ub
SOC2<1%     -0.8832  0.2934  -3.0106  0.0026  -1.4582  -0.3082   **
SOC21%~2%   -1.4887  0.3827  -3.8901  0.0001  -2.2388  -0.7387  ***
SOC2>2%     -1.3039  0.2843  -4.5871  <.0001  -1.8611  -0.7468  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

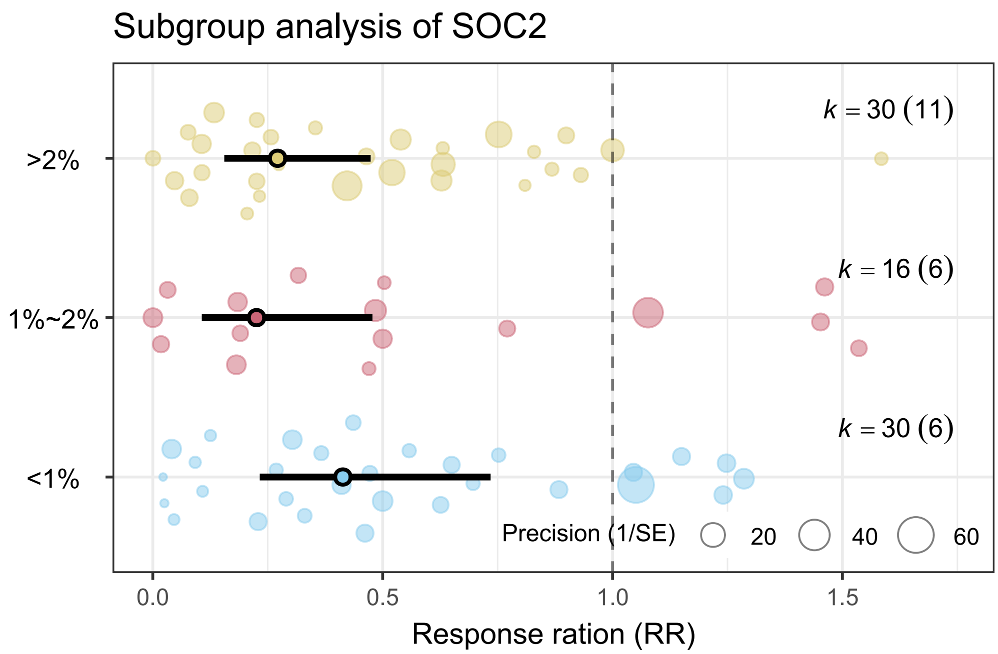
```
mod_SOC2$comp
```

```
# A tibble: 3 × 6
  level1 level2   diff    se      z     p
  <chr>  <chr>   <dbl> <dbl>  <dbl> <dbl>
1 <1%    1%~2%   0.605 0.482  1.26  0.209
2 <1%    >2%     0.421 0.408  1.03  0.303
3 1%~2%  >2%    -0.185 0.477 -0.388 0.698
```

```
mod_SOC2$mod_res
```

```
   name estimate lowerCL upperCL lowerPR upperPR
1   <1%    0.413   0.233   0.735   0.020   8.551
2 1%~2%    0.226   0.107   0.478   0.011   4.848
3   >2%    0.271   0.156   0.474   0.013   5.596
```

```
mod_SOC2$fig
```



7.15 Intial soil moisture

```
mod_SWC <- sub_meta(dat = es, mod = "SWC_w",
                         transfm = "none", es.lab="ln(RR)")
mod_SWC$model
```

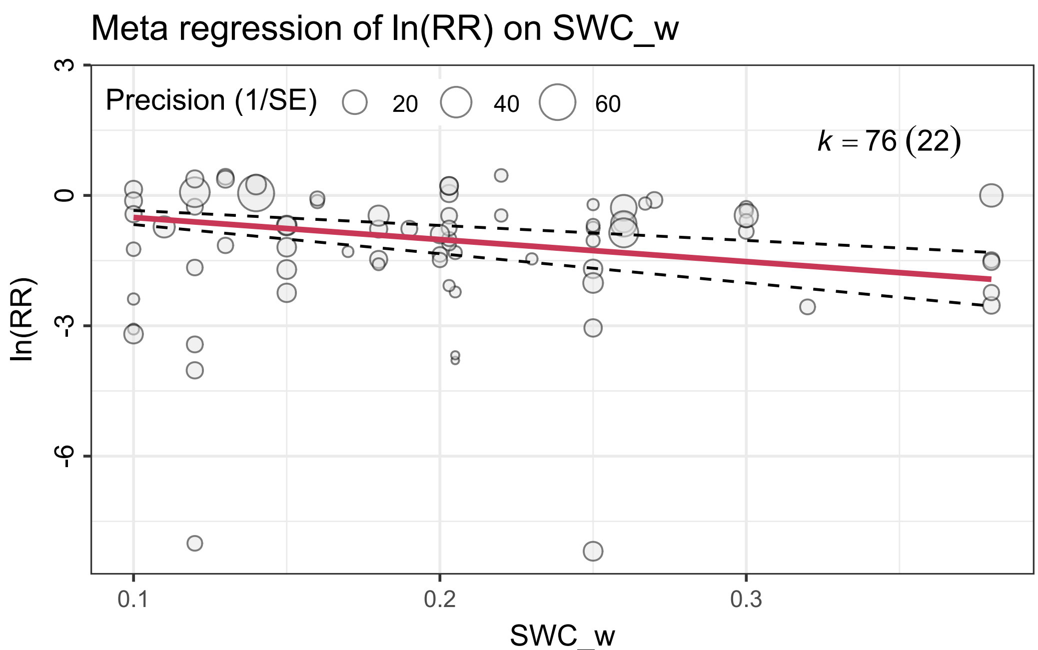
```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.3160 (SE = 0.3998)
tau (square root of estimated tau^2 value):             1.5218
I^2 (residual heterogeneity / unaccounted variability): 99.67%
H^2 (unaccounted variability / sampling variability):   304.27

Test for Residual Heterogeneity:
QE(df = 75) = 14151.0452, p-val < .0001

Model Results:

       estimate      se      zval     pval     ci.lb     ci.ub
SWC_w   -5.0771  0.8285  -6.1280  <.0001  -6.7009  -3.4532  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_SWC$comp
```

```
Parameter1 | Parameter2 |    r |        95% CI | t(74) |     p
---------------------------------------------------------------
SWC_w      |         yi | 0.02 | [-0.21, 0.24] |  0.15 | 0.882

Observations: 76
```

```
mod_SWC$fig
```

```
Ignoring unknown labels:
• parse : "TRUE"
```




```
mod_SWC2 <- sub_meta(dat = es, mod = "SW_condition",
                     transfm = "exp", es.lab="Response ration (RR)")
mod_SWC2$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2752 (SE = 0.3958)
tau (square root of estimated tau^2 value):             1.5084
```

```
I^2 (residual heterogeneity / unaccounted variability): 99.64%
H^2 (unaccounted variability / sampling variability):   278.61

Test for Residual Heterogeneity:
QE(df = 74) = 10972.3877, p-val < .0001

Test of Moderators (coefficients 1:2):
QM(df = 2) = 45.9496, p-val < .0001

Model Results:

                 estimate      se     zval    pval    ci.lb    ci.ub
SW_conditiondry   -1.7131  0.4067  -4.2127  <.0001  -2.5101  -0.9161  ***
SW_conditionwet   -1.0593  0.1995  -5.3106  <.0001  -1.4503  -0.6683  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

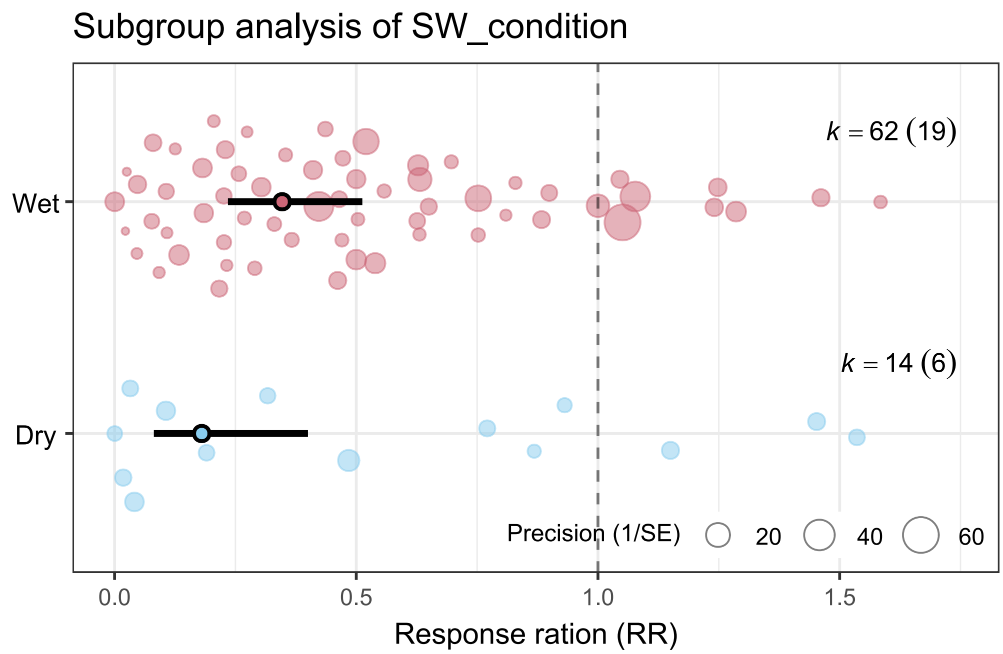
```
mod_SWC2$comp
```

```
# A tibble: 1 × 6
  level1 level2   diff    se     z     p
  <chr>  <chr>   <dbl> <dbl> <dbl> <dbl>
1 dry    wet    -0.654 0.453 -1.44 0.149
```

```
mod_SWC2$mod_res
```

```
  name estimate lowerCL upperCL lowerPR upperPR
1  Dry    0.180   0.081   0.400   0.008   3.853
2  Wet    0.347   0.235   0.513   0.018   6.840
```

```
mod_SWC2$fig
```



7.16 Air temperature

```
mod_air.temp <- sub_meta(dat = es, mod = "air_temp",
                         transfm = "none", es.lab="ln(RR)")
mod_air.temp$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2782 (SE = 0.3935)
tau (square root of estimated tau^2 value):             1.5094
I^2 (residual heterogeneity / unaccounted variability): 99.64%
H^2 (unaccounted variability / sampling variability):   278.10

Test for Residual Heterogeneity:
QE(df = 75) = 10946.4601, p-val < .0001

Model Results:

          estimate      se     zval    pval    ci.lb    ci.ub
air_temp   -0.0474  0.0086  -5.5128  <.0001  -0.0643  -0.0306  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

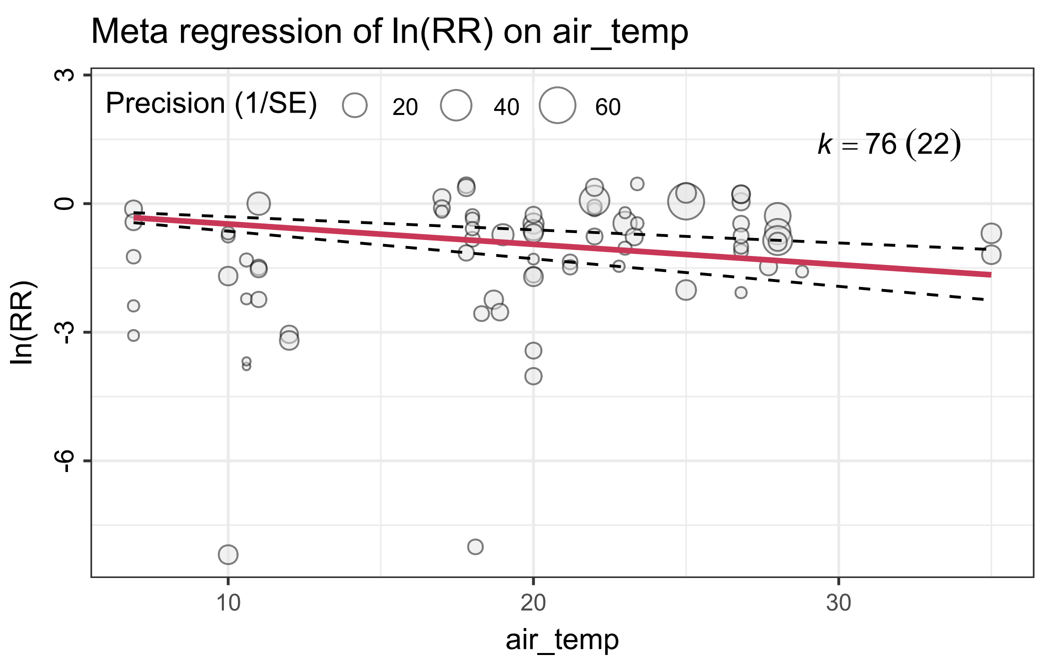
```
```

```r
mod_air.temp$comp
```

```
Parameter1 | Parameter2 |    r |       95% CI | t(74) |       p
----------------------------------------------------------------
air_temp   |         yi | 0.32 | [0.10, 0.51] |  2.86 | 0.005**

Observations: 76
```

```r
mod_air.temp$fig
```

```
Ignoring unknown labels:
• parse : "TRUE"
```



7.17 Rainfall

```r
mod_rainfall <- sub_meta(dat = es, mod = "rainfall",
                         transfm = "none", es.lab="ln(RR)")
mod_rainfall$model
```

```

Mixed-Effects Model (k = 76; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity):     2.2928 (SE = 0.3958)
```

```
tau (square root of estimated tau^2 value):             1.5142
I^2 (residual heterogeneity / unaccounted variability): 99.68%
H^2 (unaccounted variability / sampling variability):   307.88

Test for Residual Heterogeneity:
QE(df = 75) = 15222.7705, p-val < .0001

Model Results:

          estimate      se     zval    pval    ci.lb    ci.ub
rainfall   -0.0096  0.0030  -3.2170  0.0013  -0.0155  -0.0038  **

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_rainfall$comp
```

```
Parameter1 | Parameter2 |    r |        95% CI | t(74) |     p
--------------------------------------------------------------
rainfall   |         yi | 0.13 | [-0.09, 0.35] |  1.16 | 0.250

Observations: 76
```

```
mod_rainfall$fig
```

```
Ignoring unknown labels:
• parse : "TRUE"
```

Meta regression of ln(RR) on rainfall